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Probing temperature able to improve Raman spectroscopic discrimination of adulterated olive oils

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ABSTRACT

An effective method for improving Raman spectroscopic identification of adulterated extra virgin olive oil (EVOO) was proposed. It was based on probing an optimal sample temperature where the spectral differentiation among components was enhanced, since temperature-induced spectral variations of individual components expected to be diversely different depends on their molecular structures. Initially, oleic and linoleic acids were separately frozen, and Raman spectra were consecutively collected at regular intervals as the temperature increased. The spectral variations of oleic and linoleic acids were dissimilar with each other due the difference in their molecular structures. In the temperature-varied spectra of pure EVOOs and adulterated EVOOs, both the magnitudes and patterns of the spectral differences between the two groups differed, and the measurement at −36.4 °C resulted in the lowest discrimination error of 8.0% based on linear discriminant analysis using principal component scores. The spectral features of the two groups became more discernible at this temperature due to the dissimilar temperature-dependent molecular vibrations of the components, which improved the authentication accuracy.
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1. Introduction

Near-infrared (NIR), infrared (IR), and Raman spectroscopy have been studied for authentication of edible oils, especially olive oils [1–8]. These methods are comparable for such authentication; while, the most intrinsic way to improve the accuracy is enhancement of the spectral differentiation between pure oil and adulterants. In this regard, one simple approach to improve the authentication would be probing an optimal measurement temperature leading to enhanced spectral discrimination among components. Since the temperature-dependent structural variations of components are expected to differ, individual components could be more structurally discernible at a certain stage and the use of spectra collected under this condition could improve discrimination. As the major components of olive oils are fatty acids possessing long alkyl chains with different degrees of unsaturation, their vibrations could easily be influenced by temperature and phase changes, and these temperature-induced spectral features should differ according to the molecular structures of the components.

Here, sample temperatures in the sub-zero temperature range yielding enhanced Raman spectroscopic differentiation between EVOO and an adulterant (soybean oil) have been searched to ultimately improve the accuracy of authentication. For preliminary examination, oleic and linoleic acids (the main component in EVOO and soybean oil, respectively) were frozen in liquid nitrogen and then simply exposed to ambient conditions to naturally increase the sample temperature. During the temperature elevation, Raman spectra were consecutively collected at regular intervals, and the temperature-dependent spectral variations of these two samples were compared. Next, Raman spectra of pure and adulterated EVOOs containing soybean oil (degree of adulteration: 5%) were collected during the temperature elevation. The differences in the temperature-varied spectral features of pure and adulterated EVOO were examined. Then, the principal component scores obtained from the collected spectra were used for authentication through linear discrimination analysis (LDA), and the optimal temperature providing the most accurate discrimination was determined. Finally, the justification for the improved accuracy was discussed in light of the corresponding spectral features.

2. Experimental section

2.1. Preparation of samples

The EVOOs and soybean oils used in this study were purchased at a local market. Fifteen EVOOs produced by five different manufactures (3 bottles from each manufacturer) and 15 soybean oils made by three manufacturers (5 bottles from each manufacturer) were used. Thirty adulterated EVOOs were prepared through the blending of one pure EVOO and one soybean oil without duplicated mixing. The

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concentration of soybean oil in all adulterated EVOOs was 5 wt%. EVOO contains 8.0–15.0% palmitic acid, 65.0–80.0% oleic acid, 5.0–11.0% linoleic acid, and 0.5–1.5% linolenic acid [4,9–11]. Soybean oil is composed of 7.0–11.0% palmitic acid, 22.0–34.0% oleic acid, 43.0–56.0% linoleic acid, and 6.0–10.0% linolenic acid.

2.2. Acquisition of Raman spectra and data processing

Raman spectra were collected with a wide-area illumination (WAI) scheme (excitation: 785 nm), providing a laser illumination diameter of 6 mm (power: 200 mW) to cover a large sample area (28.4 mm^2) [12]. A sample (1.6 mL) was transferred into a rectangular quartz cuvette and then the sample-contained cuvette was immersed in liquid nitrogen for 7 min for complete freezing. The cuvette was then removed from the liquid nitrogen and immediately positioned at the focal point for spectral collection. The duration of each snapshot was 20 s, and a total of 50 snapshots (spectra, resolution: 4 cm^{-1}) were collected for each sample. Fig. S1 displays the actual temperature variations of the EVOO, soybean oil, and 5 wt% soybean-oil-adulterated EVOO samples during the spectral collection (Supplementary data). The sample temperature was directly recorded by inserting a thermocouple (DE-3008A T-type, Digital Thermometer, Taiwan) into the sample-contained cuvette. Baseline correction, normalization, principal component analysis (PCA) and LDA were performed with MATLAB version 7.0 (Math Works Inc., MA, USA).

3. Results and discussion

3.1. Spectral variations of pure components during temperature change

Initially, the normalized Raman spectra (1800–800 cm^{-1} range) of oleic, linoleic, and linolenic acids at room temperature were examined, as shown in Fig. 1. For the normalization, the baseline-corrected spectra at seven different wavenumbers (1800, 1700, 1550, 1390, 1168, 995, and 800 cm^{-1}) were divided by the corresponding peak area under the 1800–800 cm^{-1} range. Four major bands at 1658, 1440, 1300, and 1265 cm^{-1} are resulted from the stretching of C═C bonds, scissoring deformation of CH_2 groups, in-phase methylene twisting, and in-plane C═C—H deformation, respectively [1,2,6,8,13,14]. Since the peaks at 1658 and 1265 cm^{-1} are originated from the vibrations of double bonds, the peak intensities increase with the degree of unsaturation and thus have the highest intensity for linolenic acid. Conversely, the intensities of the peaks at 1440 and 1300 cm^{-1}, originated from the vibrations of aliphatic structures, decrease as the number of double bonds increases. Meanwhile, the differences in band shapes are more characteristic in the 1180–800 cm^{-1} range, where the backbone vibrations of C—C bonds and stretching of C—O bonds appear.

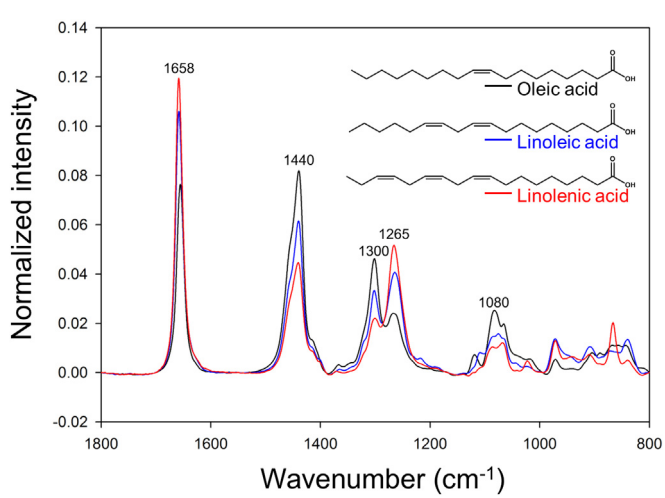

Fig. 1. Raman spectra of oleic, linoleic, and linolenic acids, measured at room temperature.

Next, Raman spectra of oleic (T_m: 16.3 °C) and linoleic acids (T_m: −6.9 °C) were collected by changing the sample temperature approximately from −169.9 to 13.8 °C. Fig. 2(a) and (b) displays the Raman spectra (1800–1600 cm^{-1} range) of oleic and linoleic acids, respectively, measured at the 1st, 5th, 10th, 15th, 20th, 25th, and 50th snapshots. The spectra collected at room temperature (dashed line) are also displayed. First, the position of the band (C═C stretching) is shifted to the higher wavenumber at the 1st snapshot and then moves to the lower wavenumber as the temperature increases. The frozen state at lower temperatures makes the C═C vibration more rigid, causing the band to shift to the higher wavenumber. Second, the peak intensity increases as the temperature increases. The less condensed surroundings at higher temperatures make the C═C bonds more deformable (polarizable), thus increasing the intensity. As found, the magnitudes of the band shift as well as intensity change are different.

Fig. 2(c) and (d) shows the same sample spectra in the 1520–1200 cm^{-1} range. The intensities of the 1440 cm^{-1} (scissoring deformation of CH_2 groups) and 1300 cm^{-1} (in-phase methylene twisting) bands are strongest in the 1st snapshot in both cases. At a temperature far below the freezing points of these fatty acids, the hydrocarbon backbones are stiffened and the molecules become more densely arranged, thus resulting in the increase intensity. The rigid structure and dense arrangement of the molecules relax as the temperature increases, so the intensity decreases and the detailed vibrational bands fade at the 50th snapshot in both cases. Meanwhile, linoleic acid exhibits a greater degree of temperature-induced spectral variation than oleic acid.

For both samples, the peaks in the 1200–800 cm^{-1} range are much narrower and more distinct at lower temperatures, as shown in Fig. 2(e) and (f). For example, the appearance of new narrow bands of linoleic acid at 1040 and 1002 cm^{-1} is notable (marked by asterisks). Again, the temperature-induced structural variation of linoleic acid is more significant and more diverse than that of oleic acid. Overall, during temperature change, the spectral variations of oleic and linoleic acids are fairly dissimilar over the whole spectral range.

3.2. Spectral variations of EVOO and soybean oil during temperature change

Fig. 3(a) displays the Raman spectra of all EVOOs (blue), soybean oils (red), and 5 wt%-soybean-oil-adulterated EVOOs (green, hereafter referred to as 5%-adulterated EVOOs) measured at room temperature. Noticeably, the band at 1750 cm^{-1} corresponding to the stretching of C═O in glyceryl esters was observed; while, it was absent in the spectra of the fatty acids. Since the concentration of oleic acid is higher in EVOO than in soybean oil, the intensities of the 1658 and 1265 cm^{-1} peaks associated with the vibrations of double bonds are lower in EVOO, while the intensities of the aliphatic peaks at 1440 and 1300 cm^{-1} are greater, as expected. The 1180–800 cm^{-1} range exhibits more dissimilar features between EVOO and soybean oil. The 5% adulteration of EVOO by soybean oil did not significantly change the overall spectral features of EVOO, due to the low adulteration rate. For detailed examination, the 1320–1254, 1126–1012, and 990–800 cm^{-1} ranges are separately highlighted in Fig. 3(b), (c), and (d), respectively. As expected, the Raman spectra of EVOOs and 5%-adulterated EVOOs overlap considerably with each other over the spectral ranges. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Since oleic acid and linoleic acid are the main components of EVOO and soybean oil, respectively, the temperature-induced spectral variations of EVOO and soybean oil expect to be analogous to

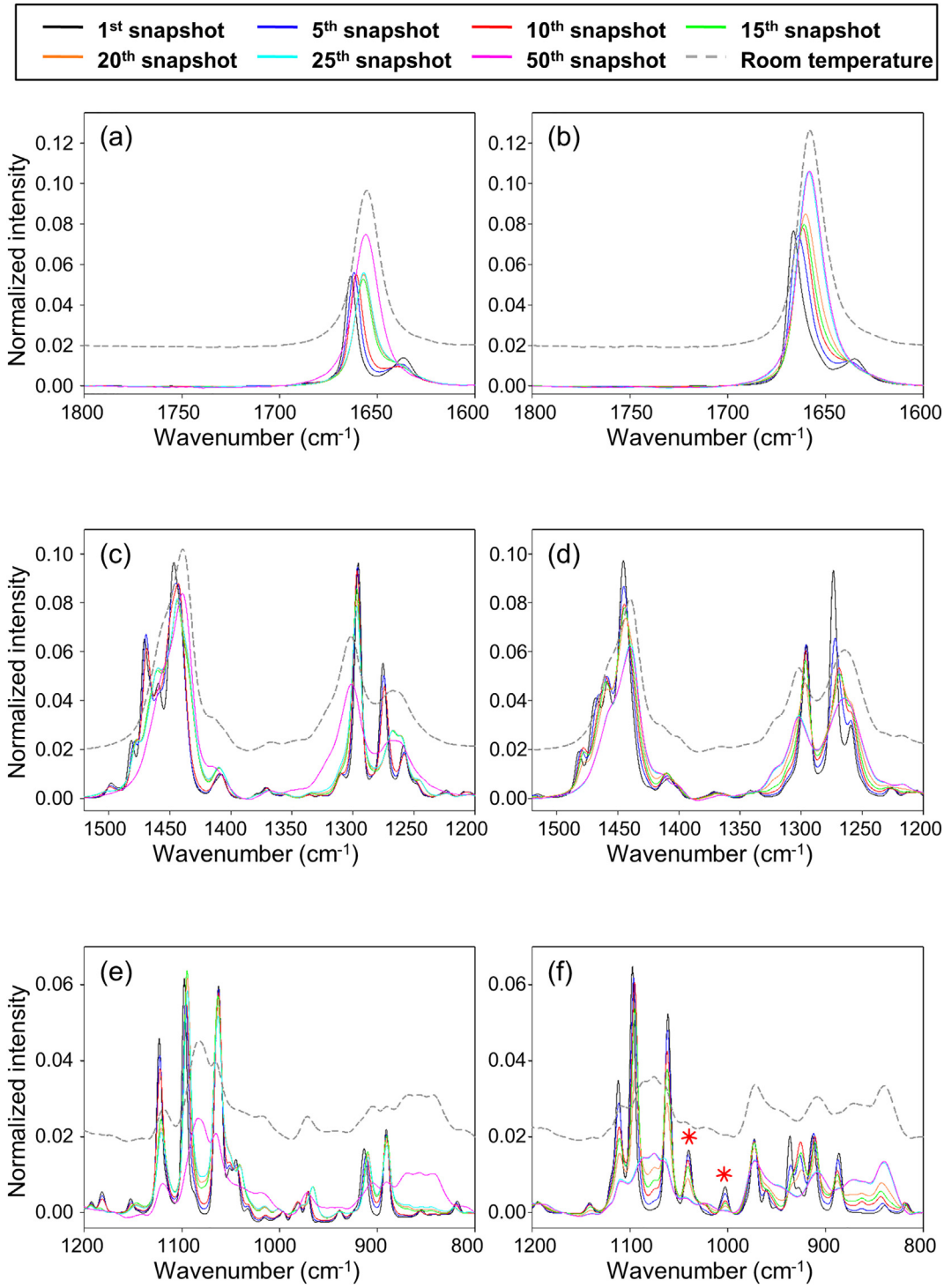

Fig. 2. Raman spectra of oleic acid (left column) and linoleic acid (right column) collected during temperature elevation. The 1800–1600 ((a), (b)), 1520–1200 ((c), (d)), and 1200–800 ((e), (f)) cm^{-1} ranges are highlighted. New narrow bands of linoleic acid at 1040 and 1002 cm^{-1} are marked by asterisks.

those of oleic acid and linoleic acid, respectively. Fig. 4 shows Raman spectra of EVOO and soybean oil acquired with the temperature elevation, in the same fashion described above. The spectra of EVOO and soybean oil are shown in the left and right columns, respectively, and the 1800–1600 cm^{-1} ((a) and (b)), 1520–1200 cm^{-1} ((c) and (d)), and 1200–800 cm^{-1} ((e) and (f)) ranges are highlighted separately. The dashed-line spectra acquired at room temperature are also shown (arbitrarily offset for clarity). The shifts of band position, intensity changes, and band shape variations resulting from the temperature elevation are apparent and are generally similar to those observed in the spectra of oleic and linoleic acids (refer to Fig. 3).

For the investigation on the degree of spectral discrimination according to the temperature change, difference spectra were obtained through the subtraction of the pure EVOO spectrum from the 5%-

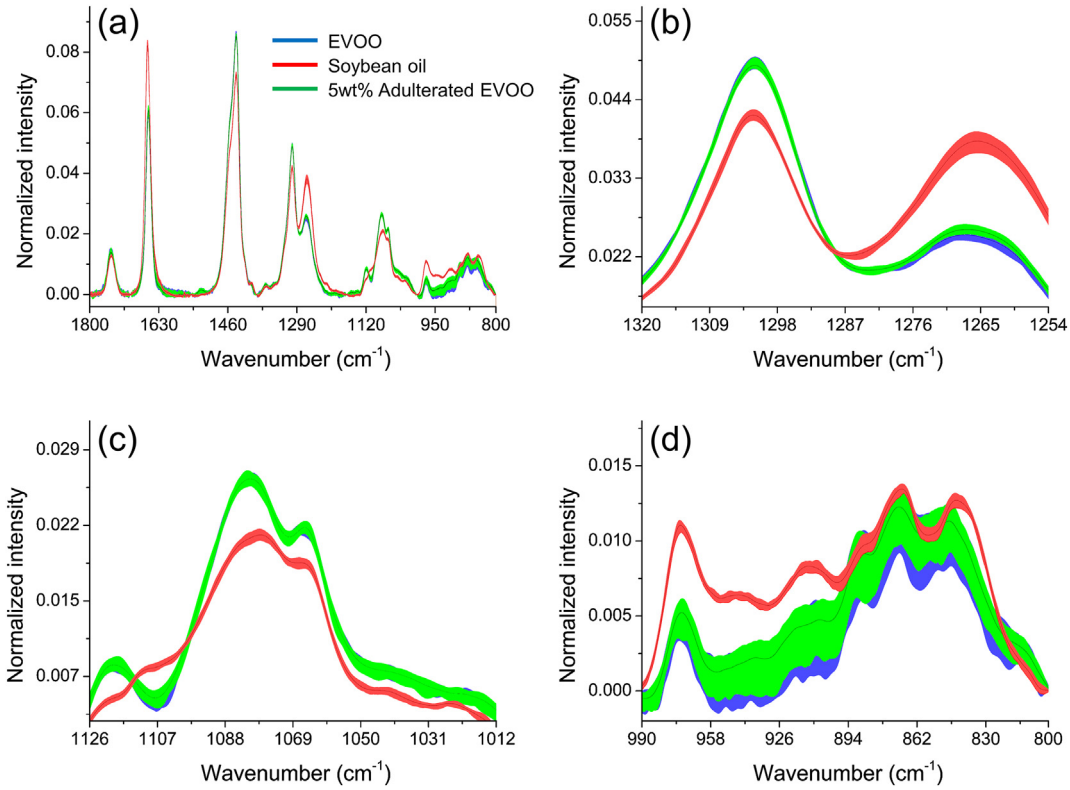

Fig. 3. Raman spectra (1800–800 cm^{-1}) of all EVOOs (blue), soybean oils (red), and 5%-adulterated EVOOs (green), measured at room temperature are shown (a). The 1320–1254 (b), 1126–1012 (c), and 990–800 cm^{-1} (d) ranges are highlighted separately. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

adulterated EVOO spectrum at each snapshot, as shown in Fig. 5. The sample temperature corresponding to each snapshot is also indicted. The difference features at the low temperatures (i.e., in the 1st through 25th snapshots) are more characteristic than those at room temperature. The differences in the band intensities and shapes are apparent over the spectral range, and their features change as the temperature varies. Thus, the discrimination could be potentially improved at certain stages.

3.3. Principal component analysis (PCA) and linear discriminant analysis (LDA)

Using PCA-LDA, the discrimination accuracy at room temperature measurement was initially estimated. The PCA scores obtained from the spectra in the 1800–800 cm^{-1} range were fed for LDA. Fig. 6(a) shows the variation of error according to the increase the number of employed scores for LDA. The error was obtained by cross-validation of the whole dataset (60 samples, 30 EVOOs and 30 adulterated EVOOs) 200 times with random selections of the training (48 samples, 24 EVOOs and 24 adulterated EVOOs) and validation (12 samples) samples at each cross-validation. As typically observed, the error decreases with the incorporation of more scores and becomes minimized to 13.7% by using the first five scores. No further accuracy improvement is observed afterward.

Next, the discrimination errors at each snapshot measurement were also obtained with the same fashion above. Fig. 6(b) shows the errors at each snapshot in the case of using the first two, three, four and five scores for LDA. The sample temperature corresponding to each snapshot is also indicated on the upper *x*-axis. Since the errors resulted from using only the first score were substantially higher than the others, they were not shown here for easy comparison of the results. Generally, the errors decrease with the incorporation of more scores; while, the error is lowest 8.0% at the 11th snapshot (−36.4 °C) with the use of first three scores. Also, the error at 17th snapshot (−12.8 °C) is low, such as 8.8%, respectively. The use of extra scores such as totals of four and five scores did not improve the accuracy.

In the room temperature measurement, the percentage of spectral variation described by the first five scores was 88.5%, while 79.3% using the first three scores in the 11th snapshot measurement. Although only 79.3% of total spectral variation in the 11th snapshot was reflected for the discrimination, the accuracy was improved. This analysis clearly demonstrates that the more characteristic and distinct spectral differences easily recognizable by PCA are present around this snapshot as shown in Fig. 5, although it would be hard to specifically pinpoint the origin of the lowest error at the 11th snapshot. In comparison with the error at room temperature measurement, the accuracy improvement was statistically significant at the 99.9% confidence level based on *t*-test ($t_{calculated}$: 8.301 > t_{table}: 3.291). Again, the dissimilar temperature-dependent structural variations of individual components, providing more discernible spectral features between the two groups, are responsible for the improved discrimination.

4. Conclusions

Determining the temperature that maximizes spectral differentiation among components was a simple and effective way to improve the accuracy of discrimination between different sample groups, such as the EVOOs and adulterated EVOOs. In contrast to routine

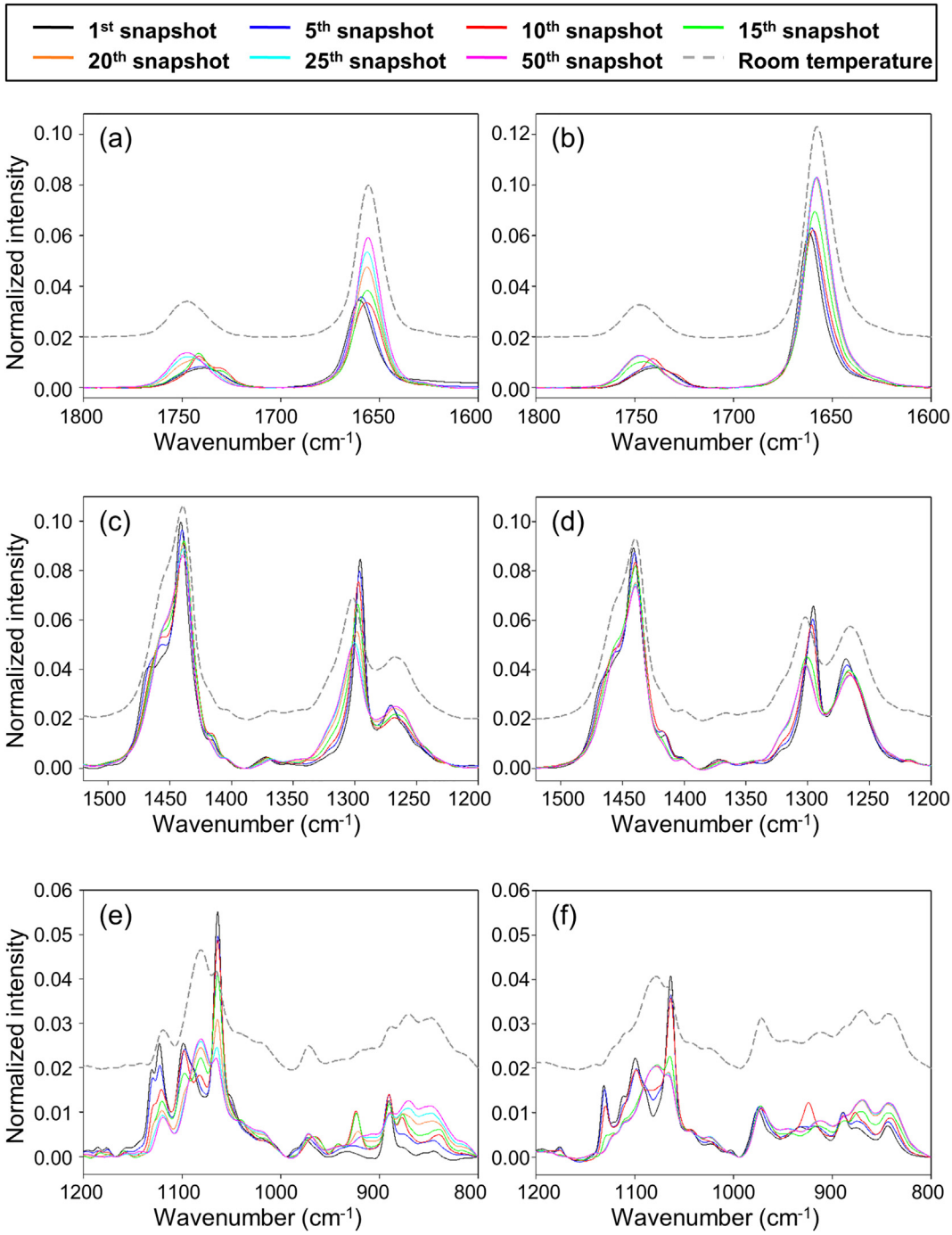

Fig. 4. Raman spectra of EVOO (left column) and soybean oil (right column) collected during temperature elevation. The 1800–1600 ((a), (b)), 1520–1200 ((c), (d)), and 1200–800 ((e), (f)) cm^{-1} ranges are highlighted.

sample measurement at room temperature, the demonstrated scheme was versatile enough to improve the discrimination accuracy without requiring extensive peripheral systems. In future, for a practical field application of the proposed method, it is necessary to prepare more samples from widely different producers to incorporate the maximized compositional variation in a calibration dataset. With this robust dataset, it will be worthwhile to evaluate other potential discriminators such as *k*-nearest neighbor (*k*-NN) and support vector machine (SVM), since their characteristics in discrimi nation are different from LDA. Whichever discrimination method is chosen, accompaniment with the demonstrated strategy of enhancing the spectral differentiation at an optimal temperature will be useful to improve discrimination accuracy further.

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.microc.2017.06.002.

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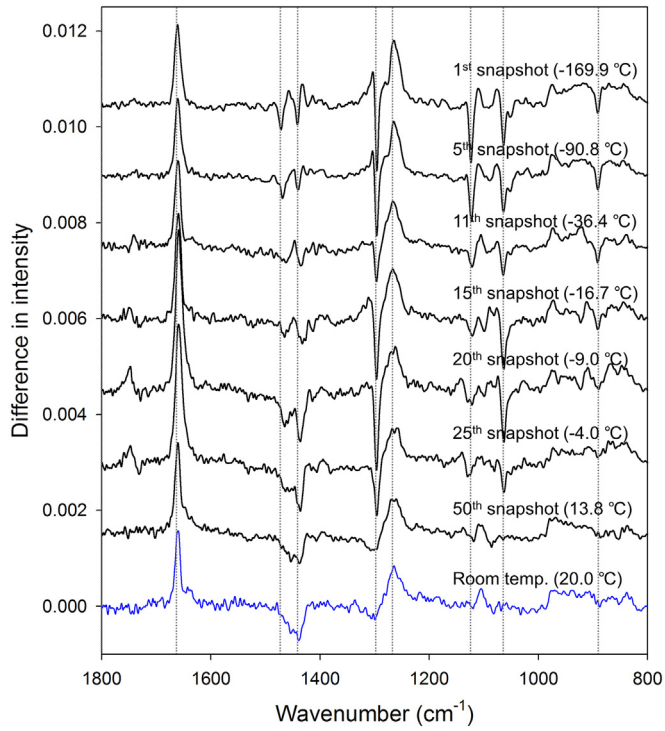

Fig. 5. Difference spectra obtained through the subtraction of the pure EVOO spectrum from the 5%-adulterated EVOO at the 1st, 5th, 11th, 15th, 20th, 25th, and 50th snapshots, as well as at room temperature. The sample temperature corresponding to each snapshot is also indicated.

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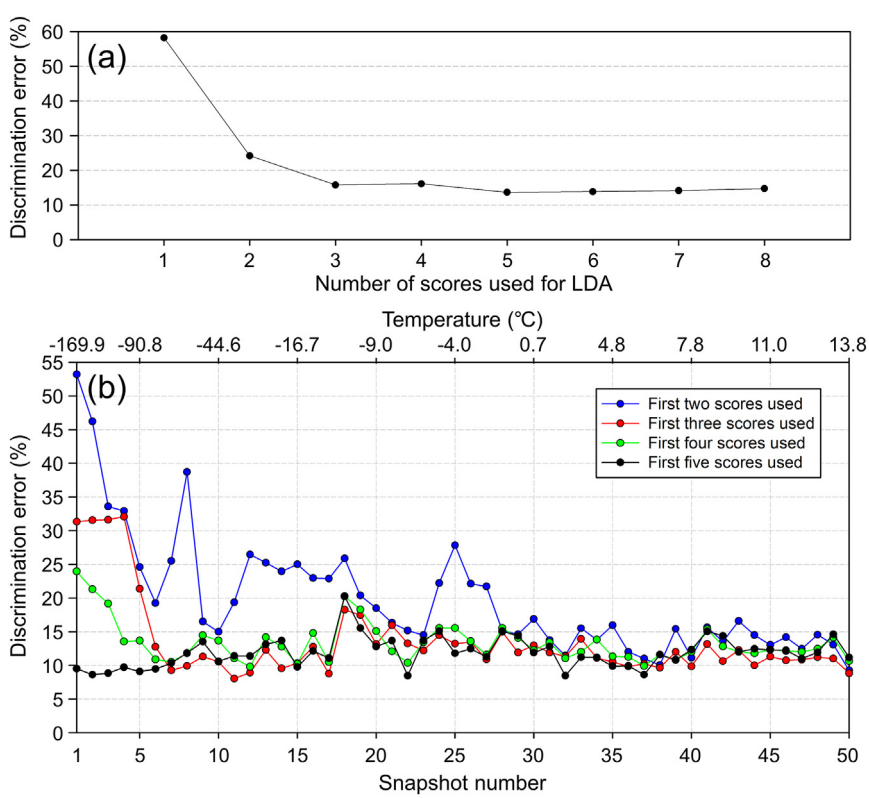

Fig. 6. Variations of discrimination errors at room temperature (a) and each snapshot measurement (b), when the numbers of scores used for LDA increase.